

Generation Interconnection Request System Impact Study Report

For Queue Project Q23-2350

Mabelvale 138 kV
September 25, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2350. The Interconnection Customer proposes a 228.6 MW battery energy storage facility interconnecting at the Mabelvale 138 kV substation in the New Jersey area. The request was received on September 21, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 356 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,921,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Carberry - Turley Substation 138 KV L
Contingency: N Albertville - Hidroelectrica Garzas 1 138 KV line
Loading: 108.5% (Rating: 247 MVA)
DFAX: 55.7%
Pre-project loading: 56.9%
2. Facility: Whiawa - N Allegheny Windpower Project 345 KV L
Contingency: Merret - Sunrise Ctr 500 KV Line
Loading: 103.6% (Rating: 1,717 MVA)
DFAX: 5.9%
Pre-project loading: 102.8%
3. Facility: Blackrock - Eklutna 345 KV
Contingency: Piney Crk Project - Caldwell 345 KV
Loading: 102.3% (Rating: 1,380 MVA)
DFAX: 5.3%
Pre-project loading: 101.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Carberry - Turley 138 KV	Reconductor line	\$962,000	100% (\$962,000)
Whiawa - N Allegheny Windpower Project 345 KV L	Reconductor line	\$9,775,000	3% (\$260,000)
Blackrock - Eklutna 345 KV	Reconductor line	\$24,164,000	3% (\$699,000)
	Total allocated to project		\$1,921,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.